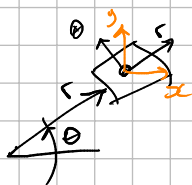


Ec. de Stokes para fluidos newtonianos

$$\tau_{ij} = -p \cdot \delta_{ij} + 2\mu V_{ij} - \frac{2}{3}\mu V_{kk} \delta_{ij}$$

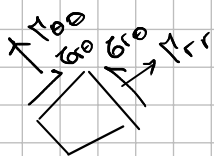
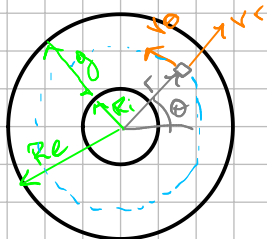


$\nabla \cdot \underline{V}$
al ser irrotacional

$$\tau_{r\theta} = -p \cdot \delta_{r\theta} + 2\mu V_{r\theta} - \frac{2}{3}\mu (V_{rr} + V_{\theta\theta} + V_{zz}) \cdot \delta_{r\theta}$$

$$\tau_{r\theta} = 2\mu V_{r\theta}$$

$$V_{r\theta} = \frac{1}{2} \left(\frac{1}{r} \frac{\partial v_r}{\partial \theta} + \frac{\partial v_\theta}{\partial r} - \frac{v_\theta}{r} \right)$$



$$\frac{\partial v_r}{\partial \theta} = 0$$

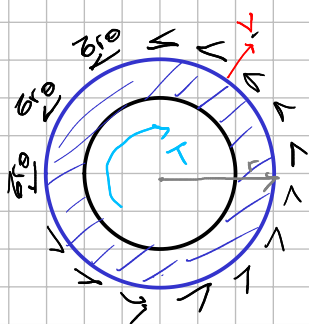
$$R_i \gg \eta \Rightarrow \frac{v_\theta}{r} \approx 0$$

$$V_{r\theta} = \frac{1}{2} \frac{\partial v_\theta}{\partial r}$$

$$\tau_{r\theta} = 2\mu \cdot \frac{1}{2} \cdot \frac{\partial v_\theta}{\partial r} = \mu \cdot \frac{\partial v_\theta}{\partial r}$$

Ec. de Newton para fluidos

$$\tau = \mu \cdot \frac{dv_x}{dy}$$



$$-\oint_S \underline{r} \times \underline{\tau} \, ds + T = 0$$

$$\left(\underline{r} \times \underline{\tau} \right) ds = |\underline{r}| \cdot \left| \underline{\tau} \right| \sin \theta \cdot ds$$

componen-
tangencial



$$T = \int_{z=0}^{z=L} \int_{\theta=0}^{\theta=2\pi} r \cdot dF = \int_0^L \int_0^{2\pi} r \cdot \tau_{r\theta} \cdot r \, d\theta \cdot dz =$$

$$= \tau_{r\theta} \cdot r^2 \cdot 2\pi L \Rightarrow \tau_{r\theta} = \frac{T}{r^2 \cdot 2\pi L}$$

$$\frac{\partial v_\theta}{\partial r} = \frac{T}{2\pi r^2 L \mu}$$

$$\int_{R_i}^r \frac{\partial v_\theta}{\partial r} dr = v_\theta(r) - \underbrace{v_\theta(R_i)}_{\text{Condición de borde}} = \int_{R_i}^r \frac{T}{2\pi r^2 L \mu} dr = \frac{T}{2\pi L \mu} \left(\frac{1}{R_i} - \frac{1}{r} \right)$$

Condición de borde $\Rightarrow v_\theta(R_i) = 0$

Perfil de velocidad

$$v_\theta(r) = \frac{T}{2\pi L \mu} \left(\frac{1}{R_i} - \frac{1}{r} \right)$$



$$v_\theta(R_e) = \omega \cdot R_e = \frac{T}{2\pi L \mu} \left(\frac{1}{R_i} - \frac{1}{R_e} \right)$$

Viscosidad del fluido

$$\mu = \frac{T}{2\pi L \omega R_e} \left(\frac{1}{R_i} - \frac{1}{R_e} \right) \rightarrow \text{Ecuación}$$

$$T = \mu \cdot \frac{2\pi L \omega R_e}{\left(\frac{1}{R_i} - \frac{1}{R_e} \right)}$$

$$T = f(\omega)$$